

JOSEPH PERRELLO

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US Citizen in Santa Cruz, California

EDUCATION

- M.S in Computer Science and Engineering from UC, Santa Cruz Status - Graduated
- Cisco Certified Network Associate (CCNA)

WORK EXPERIENCE AND THESIS

M.S. Thesis: Saturn: Zero-Configuration AI Service Discovery September 2025 - April 2026

- Created AI infrastructure for my Master's thesis that broadcasts LLM services over a network so any app can find and use them instantly, the same way computers find printers.
- Deployed the protocol in production on a live university network via a home router and Rust scripting, providing interoperability across 7 consumers in 3 languages (Python, Rust, TypeScript). This allowed students inside my Lab to access the AI services I paid for free of charge.
- Designed an agentic system of headless Claude Code sessions with grader subagents that drafted, reviewed, and fact-checked thesis chapters, using multi-agent orchestration to hit a strict deadline.
- Built a Vercel AI SDK provider in TypeScript and MCP server so existing AI applications could route LLM calls to Saturn-discovered endpoints without changing application code.

Teaching Assistant in Database Systems at University of California, Santa Cruz January 2025 - January 2026

- Taught weekly SQL and database labs to 161 college students and wrote all of the study materials, lab questions, and grading code they were tested against.
- Authored the PostgreSQL and Docker lab environments used through midterms and finals so every student practiced SQL against an identical, reproducible setup.
- Communicated technical database concepts to a non-technical audience of students, with 87.5% rating assignment help and 82.5% rating concept explanation as "Very Frequently" delivered in evaluations.
- Coordinated with TAs, graders, and the professor over Slack across four 10-week terms to keep grading standards and student priorities aligned.

In Person Instructor for ID Tech Camps at Stanford University June 2025 - August 2025

- Taught kids aged 10 to 16 how to program in Python and Java, with each student finishing their own video game by the end of every week.
- Led the camp's only Artificial Intelligence course, teaching students to integrate LLMs like Claude and GPT into their personal projects which were showcased to their parents by the end of each week.
- Communicated technical details to a non-technical audience of children and parents by translating AI and programming concepts into attention grabbing, easy to understand daily demos.

REFERENCES

Name: Adam Smith, Title: Assistant Professor, Phone: 831-295-2624, Email: amsmith@ucsc.edu

Name: Kevin Gasaway, Title: Media Prod Spec, Phone: 209-667-3173, Email: kgasaway@csustan.edu